Computers and Fluids 199 (2020) 104439

Contents lists available at ScienceDirect

Computers and Fluids

journal homepage: www.elsevier.com/locate/compfluid

On shock sensors for hybrid compact/WENO schemes

Guo-Yan Zhao [a,*], Ming-Bo Sun [a], Sergio Pirozzoli [b]

[a] Science and Technology on Scramjet Laboratory, National University of Defense Technology, Deya Road 109, Changsha 410073, China
[b] Dipartimento di Ingegneria Meccanica e Aerospaziale, Sapienza Università di Roma, Via Eudossiana 18, Roma 00184, Italy

ARTICLE INFO

Article history:
Received 27 August 2019
Revised 10 December 2019
Accepted 9 January 2020
Available online 13 January 2020

Keywords:
Finite difference methods
Shock capturing
Hybrid schemes
Shock sensors

ABSTRACT

We present a systematic framework for the evaluation of shock sensors in high-resolution hybrid compact/WENO algorithms, with the goals of testing robustness and establishing accuracy in wavenumber space, with the intent to go beyond mere case-by-case comparison of solvers/sensors. Several sensors are considered, including classical ones, sensors based on multi-resolution wavelet analysis, and WENO-based shock sensors. The crucial issue of identification of suitable thresholds for the various shock sensors is tackled through a series of static tests and numerical simulations. Whereas any shock sensor with suitably low threshold is found to be effective in capturing shocks, not all of them perform equally well for waves. Performance degradation in numerical simulation of wave-like phenomena is here characterized by introducing an effective band-width for each sensor, which is found to be quite narrow for classical sensors, and much wider for WENO-based shock sensors. Based on this analysis, we also identify a new shock sensor which can be easily implemented in existing WENO-based codes. The new sensor doesn't suffer from contamination from the WENO numerical error even for waves resolved with as few as three points-per-wavelength. This property translates into improved performance in wavenumber space, and greater resolving power in flow cases involving shocks and compressible turbulence, as demonstrated through a series of numerical tests. We emphasize that this sensor can be applied more broadly to any algorithm which contains similar smoothness indicators as classical WENO.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

Hybrid shock-capturing schemes, probably first proposed by Adams and Shariff [1] and then improved by Pirozzoli [2], have been widely applied for numerical simulation of compressible turbulence as well as for computational aero-acoustics in the presence of shocks, and in general whenever high accuracy and capability of resolving a broad range of length scales are required, while retaining monotonic numerical shock transitions. Hybrid schemes follow the basic idea of activating large numerical diffusion embodied in shock-capturing schemes only where strictly needed, namely in the vicinity of discontinuities, and using more efficient linear (either explicit or compact) discretizations elsewhere. This idea subtends the need to provisionally locate discontinuities prior to the actual discretization step, and it is clear that the design of a proper shock sensor is a key ingredient for the formulation of accurate and stable algorithms.

Traditional shock sensors [such as [3–5]] rely on first- or second-order discrete derivatives of the flow variables for smoothness evaluation. Although capable of effectively tracing shocks, these sensors suffer from clear weaknesses, as first-order derivatives are typically large for high-frequency waves, and second-order derivatives are large at critical points. Hence, smooth sinusoidal waves may be erroneously interpreted as non-smooth functions, which comes at the expense of poor efficiency and accuracy. Discerning shock/discontinuities from high-frequency waves and critical points is in fact quite difficult, and it is the main subject of the present paper. Many studies have focused on this challenging topic in recent years. Bogey et al. [6] used a sensor similar to Jameson's namely based on the high-frequency content (quantified in terms of the second derivative) of pressure to switch between a second-order shock-capturing and a higher-order filter. Oliveira et al. [7] introduced a procedure based on analysis of the truncation error ratio across multiple grid resolutions, and local left- and right-hand slope ratio check. Highly sophisticated sensors based on multi-resolution wavelet analysis were developed by Sjogreen and Yee [8], which were applied to achieve shock capturing through local nonlinear filtering of high-order non-dissipative base schemes. Multi-resolution-based sensors were tested in a variety of additional publications which supported great robustness and accuracy of the approach [e.g. [9–14]]. Hill and Pullin [15] initiated the now widely used practice of

* Corresponding author.
E-mail addresses: zhaoguoyan09@nudt.edu.cn (G.-Y. Zhao), sunmingbo@nudt.edu.cn (M.-B. Sun), sergio.pirozzoli@uniroma1.it (S. Pirozzoli).

https://doi.org/10.1016/j.compfluid.2020.104439
0045-7930/© 2020 Elsevier Ltd. All rights reserved.

exploiting WENO (weighted essentially-non-oscillatory) weights as a measure of stencil regularity. In particular, their sensor is based on the ratio of the largest to the smallest WENO weight, which becomes large whenever one or more WENO sub-stencils contain discontinuities. Similarly, Visbal and Gaitonde [16] utilized the maximum squared value of the smoothness indicators to define a shock sensor, and pointed to the importance of adding buffer points around critical nodes to guarantee oscillation-free transition between zones to be handled with a shock-capturing scheme and smooth zones. Movahed and Johnsen [17] further considered the norm of the deviation of the WENO sub-stencil weights from the ideal ones as a measure of overall stencil smoothness. Recent examples of sensors based on the WENO smoothness indicators and weights include the work of Fu et al. [18] and Peng and Shen [19]. Physics-based sensors are also widely used in practice, the most prominent example being the one introduced by Ducros et al. [20], and based on relative importance of dilatation (i.e. the velocity field divergence) to vorticity, These sensors are found to work well for shock-turbulence interaction problems. Variations of the Ducros sensor were elaborated by Larsson et al. [21] and White et al. [22], although the method is clearly inapplicable in one space dimension and/or to inviscid problems [23,24].

With the exception of Ren's sensor, which relies on seamless blending of shock-capturing and linear schemes, all the above sensors rely on identification of a suitable threshold for shock eduction, which is typically a matter of subtle compromise. In fact, low thresholds yield over-diffusive representation of waves/fine flow features, whereas large values may threaten the shock-capturing capability. Although previous authors frequently provide 'suitable' threshold values, the choice is typically made on a case-by-case basis, and with a lot of empiricism. It is the main goal of the present paper to carry out a systematic analysis of established shock sensors, with main emphasis on establishing their effective resolving power in wavenumber space, once suitable upper thresholds compatible with oscillation-free shock capturing are selected. Based on the analysis herein developed, we also present a new sensor which has the advantage of achieving robust capturing of shocks while retaining quasi-spectral representation of waves. The paper is organized as follows: in Section 2 we present the essential ingredients of hybrid compact/WENO schemes, which will be used as a workhorse for the analysis; in Section 3 several common shock sensors are reviewed, and a new sensor is proposed; in Section 4 the various sensors are tested for simple one-dimensional problems, and their performance in wavenumber space is elucidated; applicative examples to the Euler equations of gas dynamics are presented in Section 5, and concluding remarks given in Section 6.

2. Hybrid compact/WENO schemes

For later use, in this section we describe the basic formulation of a hybrid compact/WENO scheme, in which a baseline compact upwind discretization (CU) is selectively replaced with a WENO reconstruction at and near discontinuities. Although several important improvements of the WENO algorithm have been proposed in recent years [e.g. [25–30]], in this paper we mainly refer to its classical form, as introduced by Jiang and Shu [31], which is implemented in most academic compressible flow codes. A fully conservative formulation is used to facilitate coupling of schemes and enable shock-capturing via Lax-Wendroff theorem. Compact schemes have several advantages over their non-compact counterparts, as discussed in detail by Lele [32]; Adams and Shariff [1]. Consider the model scalar hyperbolic initial value problem

$$\frac{\partial u}{\partial t}+\frac{\partial f(u)}{\partial x}=0, \quad u(x,0)=u_0(x), \quad -\infty<x<+\infty, \tag{1}$$

Table 1
Coefficients for seventh-order conservative compact upwind reconstruction (CU7), needed in Eq. (3).

a_{-1}	a_0	a_1	b_{-2}	b_{-1}	b_0	b_1	b_2
1/2	1	1/4	1/60	53/80	239/240	19/240	−1/240

where $u(x,\ t)$ is the transported quantity, and $f(u)$ is the associated flux, we consider a conservative finite-difference semi-discretization of Eq. (1) on a mesh with uniform spacing h and nodes $x_j=j\cdot h$, thus yielding

$$\frac{dv_j}{dt}=L_j(v)=-\frac{1}{h}\left(\hat{f}_{j+1/2}-\hat{f}_{j-1/2}\right)\approx\left.\frac{\partial f(u)}{\partial x}\right|_j, \tag{2}$$

where $v_j(t)\approx u(x_j,\ t)$, and $\hat{f}_{j+1/2}$ is a suitable numerical flux. In all examples hereafter shown, time integration is carried out by means of a classical three-stage, third-order, Runge–Kutta scheme.

Referring, with no loss of generality, to a seventh-order compact baseline discretization (CU7), with coefficients a_ℓ, b_m, given in Table 1, we consider a hybrid numerical flux [2], here shown in the left-biased version,

$$\begin{aligned}&\left(1-\psi_{j+1/2}\right)\sum_{\ell=-1}^{1}a_\ell\hat{f}_{j+1/2+\ell}+\psi_{j+1/2}\hat{f}_{j+1/2}\\&=\left(1-\psi_{j+1/2}\right)\sum_{m=-2}^{2}b_m f_{j+m}+\psi_{j+1/2}\hat{f}^{\mathrm{SC}}_{j+1/2},\end{aligned} \tag{3}$$

where $\hat{f}^{\mathrm{SC}}_{j+1/2}$ is a shock-capturing flux (e.g. the WENO7 flux), and the switch $\psi_{j+1/2}$ is designed to be unity at shocks, and zero elsewhere. Let r_j be the nodal value of any shock sensor (to be defined in the forthcoming Section), and ϵ be a pertinent threshold value, we assume $\psi_{j+1/2}=\max(\psi_j,\psi_{j+1})$, where

$$\psi_j=\begin{cases}1, & \text{if } r_j\geq\epsilon\\ 0, & \text{otherwise}\end{cases}. \tag{4}$$

Although hybrid schemes are now frequently used, issues related to coupling of the underlying components have not been thoroughly investigated, with the notable exception of the work of Larsson and Gustafsson [33]. Interestingly, they found that if either component scheme is dissipative, the coupled system is strongly stable, which supports the use of upwind shock-capturing schemes around discontinuities. In order to further limit the onset of numerical oscillations due to coupling of different schemes, we further implement the suggestion of Visbal and Gaitonde [16], and use buffer points around critical cells, by forcing $\psi_{j-1/2}=\psi_{j+3/2}=1$, whenever $\psi_{j+1/2}=1$. Although continuous blending of schemes is also possible based on Eq. (3) [5], we find that an abruptly varying switch as dictated by Eq. (4) is more computationally efficient as it limits the computationally expensive evaluation of the WENO7 flux at all points.

When $\psi_{j+1/2}=1$, Eq. (3) is made to reduce to WENO7 scheme [31], whereby the numerical flux is assumed to be a convex combination of low-order polynomial reconstructions, with weights chosen in such way as to achieve maximum formal order of accuracy, and to assign (nearly) zero weight to reconstructions on stencils containing discontinuities. As sketched in Fig. 1, the evaluation of the numerical flux encompasses γ sub-stencils, including γ points each,

$$S_\ell=\left\{x_{j-\gamma+1+\ell},\ldots,x_{j+\ell}\right\},\quad \ell=0,1,\ldots,\gamma-1, \tag{5}$$

out of the overall reconstruction stencil

$$S=\left\{x_{j-\gamma+1},\ldots,x_{j+\gamma-1}\right\}, \tag{6}$$

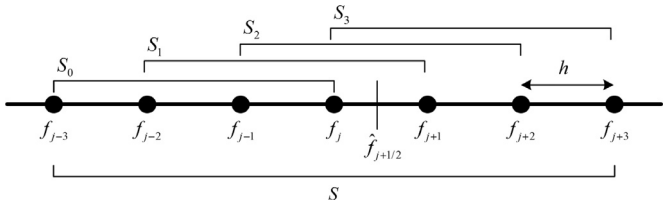


Fig. 1. Stencils for left-biased WENO7 flux reconstruction at $j+1/2$ ($\gamma = 4$).

Table 2
Ideal weights and polynomial reconstruction coefficient in WENO7 sub-stencils (see Eq. (7)).

ℓ	d_ℓ	$m=0$	$m=1$	$m=2$	$m=3$
3	4/35	1/4	13/12	−5/12	1/12
2	18/35	−1/12	7/12	7/12	−1/12
1	12/35	1/12	−5/12	13/12	1/4
0	1/35	−1/4	13/12	−23/12	25/12

which includes $2\gamma-1$ points. Let $\hat{f}^{(\ell)}_{j+1/2}$ be the reconstructed numerical flux obtained through polynomial interpolation over stencil S_ℓ,

$$\hat{f}^{(\ell)}_{j+1/2} = \sum_{m=0}^{\gamma-1} c_{\ell m} f_{j-\gamma+1+\ell+m}, \tag{7}$$

with coefficients listed in Table 2, the WENO numerical flux is evaluated as a weighted average thereof,

$$\hat{f}^{\mathrm{SC}}_{j+1/2} = \sum_{\ell=0}^{\gamma-1} \omega_\ell \hat{f}^{(\ell)}_{j+1/2}. \tag{8}$$

The weights ω_ℓ in Eq. (8) are defined as

$$\omega_\ell = \frac{\alpha_\ell}{\sum_{m=0}^{\gamma-1}\alpha_m}, \quad \alpha_\ell = \frac{d_\ell}{(\varepsilon+\beta_\ell)^2}, \quad \ell = 0,1,\ldots,\gamma-1, \tag{9}$$

where the β_ℓ are smoothness indicators for each sub-stencil, the ideal weights d_ℓ (listed in Table 2) return maximum formal order of accuracy, namely $2\gamma-1$, and $\varepsilon = 10^{-6}$ is needed to avoid divisions by zero. Expressions for the smoothness indicators for high-order WENO reconstructions were provided by Balsara and Shu [34], and are listed below for completeness for the case of WENO7 ($\gamma = 4$)

$$\begin{cases}
\beta_0 = \frac{1}{36}\left(-2f_{j-3}+9f_{j-2}-18f_{j-1}+11f_j\right)^2 + \frac{12}{13}\left(-f_{j-3}+4f_{j-2}-5f_{j-1}+2f_j\right)^2 + \frac{781}{720}\left(-f_{j-3}+3f_{j-2}-3f_{j-1}+f_j\right)^2 \\
\beta_1 = \frac{1}{36}\left(f_{j-2}-6f_{j-1}+3f_j+2f_{j+1}\right)^2 + \frac{12}{13}\left(f_{j-1}-2f_j+f_{j+1}\right)^2 + \frac{781}{720}\left(-f_{j-2}+3f_{j-1}-3f_j+f_{j+1}\right)^2 \\
\beta_2 = \frac{1}{36}\left(-2f_{j-1}-3f_j+6f_{j+1}-1f_{j+2}\right)^2 + \frac{12}{13}\left(f_{j-1}-2f_j+f_{j+1}\right)^2 + \frac{781}{720}\left(-f_{j-1}+3f_j-3f_{j+1}+f_{j+2}\right)^2 \\
\beta_3 = \frac{1}{36}\left(-11f_j+18f_{j+1}-9f_{j+2}+2f_{j+3}\right)^2 + \frac{12}{13}\left(2f_j-5f_{j+1}+4f_{j+2}-f_{j+3}\right)^2 + \frac{781}{720}\left(-f_j+3f_{j+1}-3f_{j+2}+f_{j+3}\right)^2
\end{cases}. \tag{10}$$

3. Shock sensors

Hybrid schemes require provisional detection of discontinuities in order to achieve accurate and stable algorithms. Some common shock sensors are hereafter reviewed, and a novel one is proposed.

3.1. Classical shock sensors

Harten [3] designed a shock sensor function aimed at detecting large variations in the slope of the numerical solution, defined as

$$r_j = \frac{\left|f_{j-1}-2f_j+f_{j+1}\right|}{\left|f_{j-1}-f_j\right|+\left|f_j-f_{j+1}\right|+\varepsilon}, \tag{11}$$

whereas Jameson et al. [4] considered a sensor based on the discrete local curvature of the numerical solution upon suitable normalization,

$$r_j = \frac{\left|f_{j-1}-2f_j+f_{j+1}\right|}{\left|f_{j-1}\right|+2\left|f_{j-1}\right|+\left|f_{j+1}\right|+\varepsilon}. \tag{12}$$

More recently, Ren et al. [5] considered a variation of Harten's sensor and based on consecutive discrete slopes

$$r_j = 1-\min\left(1,\frac{z_j}{r_c}\right), \quad z_j = \frac{\left|2\left(f_{j+1}-f_j\right)\left(f_j-f_{j-1}\right)\right|+\varepsilon}{\left(f_{j+1}-f_j\right)^2+\left(f_j-f_{j-1}\right)^2+\varepsilon}, \tag{13}$$

where $\varepsilon = 0.9r_c\xi^2/(1-0.9r_c)$, $\xi = 10^{-3}$, and r_c is a problem-dependent parameter. Numerical tests have shown that similar effects are obtained be deecreasing the parameter r_c as increasing the shock-capturing thrshold, ϵ. Hence, in all tests we have assumed $r_c = 0.5$. It should be noted that sensor (13) was originally used for seamless hybridization of WENO schemes with compact ones.

3.2. Multi-resolution-based shock sensors

Building on Harten's artificial compression method, Harten et al. [3] and Sjögreen and Yee [8] developed a novel type of shock sensors based on the use of redundant non-orthogonal wavelet basis functions [35,36]. The wavelet flow sensor estimates the Lipschitz exponent of a grid function f_j, as the largest χ satisfying

$$\sup\frac{\left|f_{j+1}-f_j\right|}{h^\chi} \le C, \tag{14}$$

with small or negative χ indicating poor regularity. For a C^1 wavelet function ξ with compact support, χ can be estimated from the wavelet coefficients, defined as

$$w_{m,j} = \left\langle f, \xi_{m,j}\right\rangle = \int f(x)\xi_{m,j}(x)dx, \tag{15}$$

where

$$\xi_{m,j} = 2^m\xi\left(\frac{x-j}{2^m}\right) \tag{16}$$

is the wavelet function $\xi_{m,j}$ on scale m located at point j in space.

It can be shown that if the wavelet coefficients $\max_j|w_{m,j}|$ in a neighborhood of j_0 decay as $2^{m\chi}$ as the scale is refined, then the grid function f_j has Lipschitz exponent χ at j_0. In practical computations, the smallest scale is determined by the grid size. Then $w_{m,j}$ is evaluated at the grid scale, m_0, and at a few coarser scales, say m_0+1, m_0+2, and the Lipschitz exponent at point j_0 is estimated through a least square fit [8]

$$\max_{j\,\mathrm{near}\,j_0}\log_2\left|w_{m,j}\right| = m\chi + c. \tag{17}$$

The shock sensor function should be then defined as $r_j = 1-\chi$, with a suggested shock-capturing threshold value of 0.5.

Table 3
Estimated maximum value of sensor threshold to guarantee shock capturing (ϵ_{SC}).

Sensor	Harten	Jameson	Ren	
ϵ_{SC}	0.7	10^{-3}	0.5	
Sensor	Multi-resolution	Hill	Visbal	Movahed
ϵ_{SC}	0.5	14	10^{-5}	1.1
Suggested value in Ref.	$0.4-0.6$	$20-150$	$10^{-7}-10^{-2}$	\
New sensor	$\theta=1$	$\theta=2$	$\theta=4$	
ϵ_{SC}	0.8	0.7	0.45	

3.3. WENO-based shock sensors

Many shock sensors in current use are based on the WENO smoothness indicators, so as to be able to more accurately evaluate local regularity of the numerical solution than with classical ones which merely rely on low-order derivatives. Hill and Pullin [15] proposed that the overall smoothness of the WENO stencil should be defined in terms of the ratio of the largest to the smallest smoothness indicator, which becomes very large whenever one or more WENO sub-stencils contain discontinuities, namely

$$r_j = \frac{\max_\ell \beta_\ell}{\min_m \beta_m + \varepsilon}, \quad \ell, m = 0, 1, \ldots, \gamma - 1, \tag{18}$$

Based on similar reasoning, Visbal and Gaitonde [16] considered the square the largest WENO smoothness indicator as over all sub-stencils,

$$r_j = \max_\ell {\beta_\ell}^2, \quad \ell = 0, 1, \ldots, \gamma - 1, \tag{19}$$

and Movahed and Johnsen [17] constructed a sensor based on the sum of the absolute value deviations of the nonlinear weights from their nonlinear counterparts, thus obtaining

$$r_j = \sum_{\ell=0}^{\gamma-1} |\omega_\ell - d_\ell|. \tag{20}$$

3.4. A new shock sensor

As mentioned in the introductory discussion of WENO schemes, the weight associated with a given sub-stencil (ω_ℓ) automatically reduces to the corresponding ideal weight (d_ℓ) in the case of a perfectly smooth solution, hence the difference between the two provides information about the (lack of) regularity of the solution on the sub-stencil. Similar to Movahed and Johnsen [17], It is then reasonable to judge on the regularity of the solution on the full reconstruction stencil based on a suitable norm of the distance of the WENO weights from their ideal values on all the sub-stencils, namely

$$r_j = \frac{\sum_{\ell=0}^{\gamma-1} \left| \frac{\omega_\ell}{d_\ell} - 1 \right|^\theta}{\left| \frac{1}{\min_m d_m} - 1 \right|^\theta + (\gamma - 1)}, \quad m = 0, 1, \ldots, \gamma - 1, \tag{21}$$

where the parameter θ controls the sensitivity to deviations from the ideal weights. A similar idea also subtends the sensor given in Eq. (20). The special form of the denominator in Eq. (21) is designed for normalization purposes, in such a way as to have $0 \le r_j \le 1$. Indeed, it amounts to the largest value that the numerator can ever attain, which happens when the stencil with the smallest ideal weight d_ℓ has unit weight in the final reconstruction (i.e. $\omega_\ell = 1$). Besides built-in normalization and addition of a free parameter, the main difference with respect to the sensor given in Eq. (20) is that the relative rather than the absolute deviation of the weights from their ideal values is considered, hence all sub-stencils have the same importance in defining the smoothness of the overall stencil. As we will show, this feature has some importance in the practical performance of the sensor.

4. Assessment of shock sensors for scalar equations

In the following analysis, a hybrid CU7/WENO7 scheme defined in Section 2, is used to illustrate the performance of shock sensors. Although the results of the analysis would change to some extent when choosing a different combination of baseline schemes, the procedure described above would not change. Given the use of WENO7 as a shock-capturing building block, the WENO shock sensors introduced in Sections 3.3 and 3.4 are used with $\gamma = 4$. As a preliminary step, maximum admissible values for the various sensors (ϵ_{SC}, to be used in Eq. (4)), which maintain shock-capturing capability of the resulting algorithms were identified based on extensive numerical tests, some of which are shown later on. The list of sensor thresholds is shown in Table 3, where the values suggested in the original references are also reported, and found to be generally consistent with the values that we have determined. Shock-capturing threshold values are also given for the new sensor, limited to the cases $\theta = 1, 2, 4$, and showing as anticipated that as θ increases, the upper bound decreases. Again, although the threshold values would change when using sensors based on WENO schemes of different order, the ensuing evaluation procedure would not change.

4.1. Static tests

Static tests of the various sensors have initially been carried out by monitoring their performance for assigned numerical data. In particular, a series of monochromatic test functions are essayed to characterize the spectral signature of the sensors. For the sake of illustrating the procedure, in Fig. 2 we show the computed distributions of the sensor values for an individual sinusoidal profile $f \sim e^{iwx}$, with wavenumber $w = 2\pi/\lambda$, sampled in such a way that six points-per-wavelength are used, corresponding to a reduced wavenumber $\varphi = wh = \pi/3$. As also previously noted, Harten's sensor becomes large at all extrema of the numerical solution, hence many points (specifically, those for which $r_j > 0.7$) are erroneously interpreted as shocks. A similar behavior is also happened for Jameson's, Ren's, Multi-resolution and Visbal's sensors, for which the sensor exceeds the maximum allowed threshold value at most points. The other sensors are successful in this case to distinguish the poorly resolved sine wave from a shock, and WENO is nowhere applied. However, the safety margin between the maximum value of r_j and ϵ_{SC} is very different, being a factor of about 3 for Hill's sensor, 1.5 for Movahed's sensor, and 8 for the sensor herein developed.

A complete evaluation of the sensors performance in wavenumber space is presented in Fig. 3, where we repeat the previous test for a whole range of wavenumbers, and show the maximum values of r_j over all grid points as a function of the reduced wavenumber. This representation is selected because if $\max_j r_j \le \epsilon_{SC}$, no point is handled in the WENO mode. Hence, the wavenumber diagram may be used to identify the resolving efficiency of the various

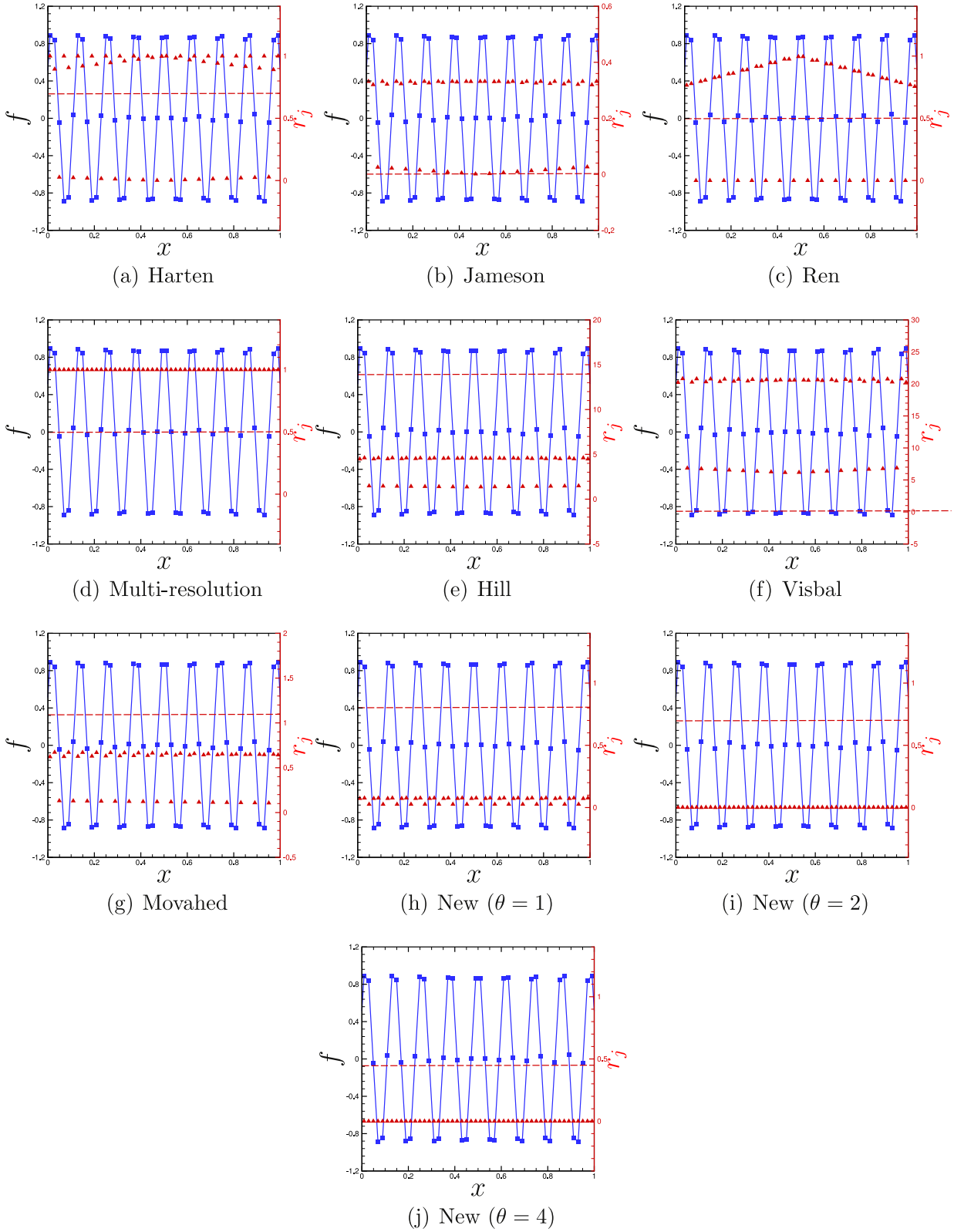


Fig. 2. Distributions of shock sensors for one-dimensional sine function f with reduced wavenumber $\varphi = \pi/3$ (square symbols). The triangle symbols (see scale on the right axis) denote values of the shock sensor. The upper threshold values for shock capturing (ϵ_{SC}, as given in Table 3) for the various sensors are marked with a dashed horizontal line.

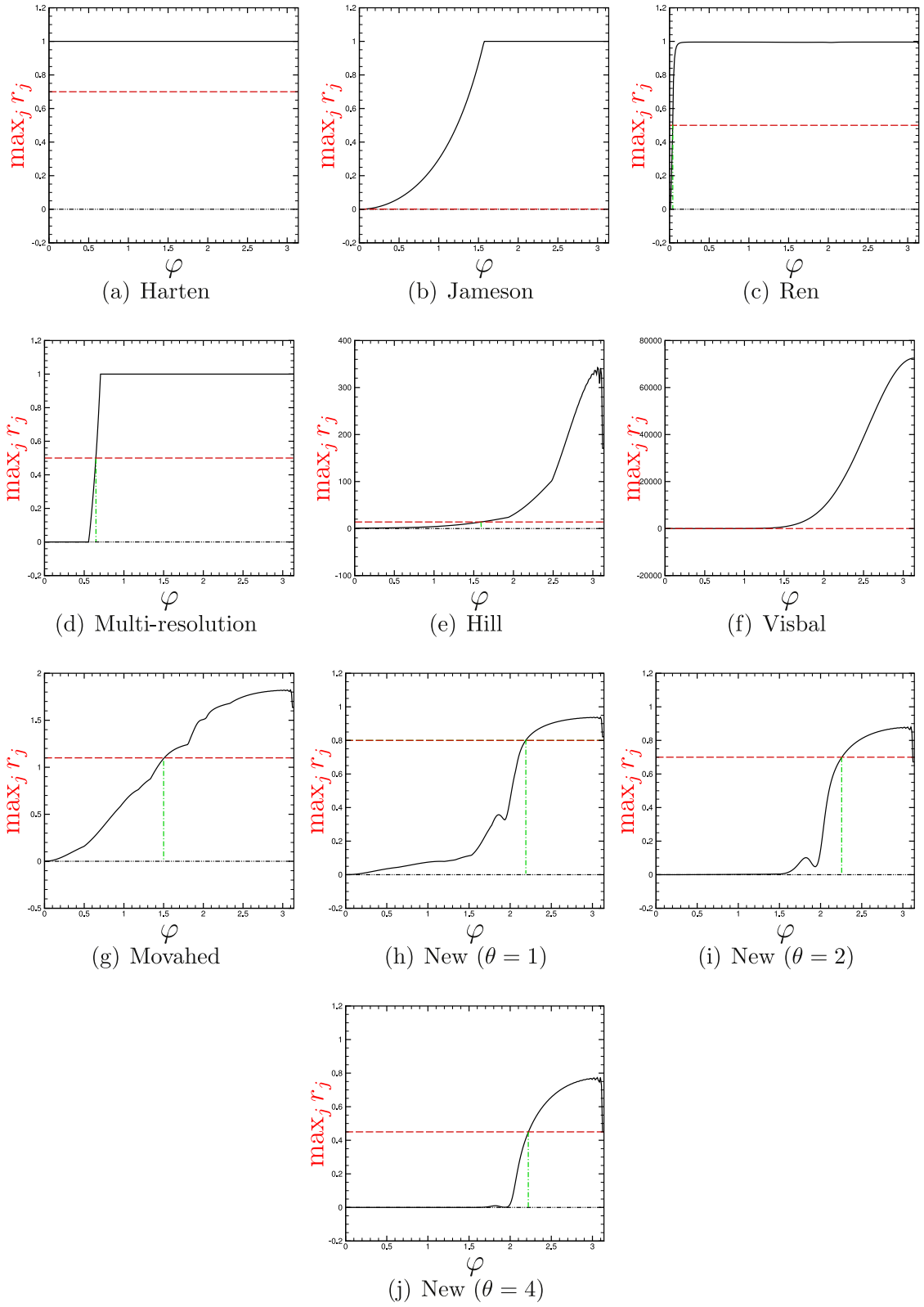


Fig. 3. Distributions of maximum value of shock sensor as a function of the reduced wavenumber, for various shock sensors. The threshold values for shock capturing (ϵ_{SC}, as given in Table 3) for the various sensors are marked with a dashed horizontal line. The vertical line marks the resolving efficiency of the various sensors.

Table 4
Resolving efficiency of various shock sensors.

Harten	Jameson	Ren	
0.0	6.32×10^{-2}	4.05×10^{-2}	
Multi-resolution	Hill	Visbal	Movahed
0.65	1.59	5.52×10^{-2}	1.50
New ($\theta = 1$)	New ($\theta = 2$)	New	
2.18	2.25	2.22	

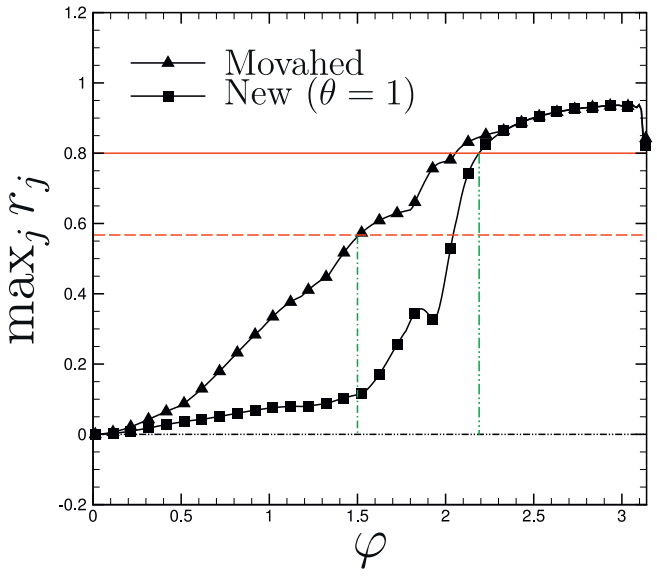


Fig. 4. Comparison of spectral signature of normalized Movahed sensor and new sensor with $\theta = 1$.

sensors, similar to what done for linear schemes by Lele [32], as the maximum value of φ for which WENO is nowhere applied, for given value of the sensor threshold. Referring to Fig. 3, the resolving efficiency may be graphically determined by looking for the intersection point of the sensor wavenumber footprint with the horizontal dashed lines indicating the threshold values. Numerical values of the resulting resolving efficiencies are provided in Table 4. It may is clearly seen that classical sensors, as well as Visbal's sensor yield a narrow range of uncontaminated wavenumbers, the Multi-resolution sensor retains accurate resolution of waves as long as at least ten points-per-wavelength are used in the discretization, Hill's and Movahed's sensors require four points-per-wavelength, and barely three points-per-wavelength suffice when the new shock sensor is used.

Given the formal similarity of shock sensor defined in Eq. (21) for $\theta = 1$ with Movahed's sensor, a comparison of their wavenumber performance is carried out in Fig. 4. For the sake of comparison, Eq. (20) is here normalized in such a way as to have $r_j \leq 1$, by defining

$$r_j = \frac{\sum_{\ell=0}^{\gamma-1} |\omega_\ell - d_\ell|}{|1 - \min_m(d_m)| + \sum_{d_n \neq \min_m(d_m)} d_n}, \quad m, n = 0, 1, \ldots, \gamma - 1. \tag{22}$$

The figure makes it clear that sensor (22) and the new sensor has very similar performance in the limit of poorly resolved waves ($\varphi \approx \pi$), which is not surprising as in that case only one of the WENO sub-stencils yields a dominant contribution to the sensor. However, the figure clearly shows that the new sensor has better wave resolving ability when $\varphi \leq 2.2$, meaning that waves with higher wavenumber may be resolved without resorting to WENO, for any given threshold set for the sensor.

4.2. Dynamic tests

As a first step toward characterization of the shock sensors when applied to compact/WENO algorithms, we have determined their empirical dispersion relation [EDR, see 37]. The EDR of a nonlinear scheme is obtained by numerically advancing sinusoidal test functions with reduced wavenumber φ up to a very short time (say τ), and determining the resulting wave amplitude through Fourier transform, say $\hat{v}(\varphi; \tau)$. This is done for all modes supported on the computational mesh to obtain a discrete representation of the spectral transfer function,

$$\Phi(\varphi) = -\frac{1}{\mathrm{i}\sigma} \log\left(\frac{\hat{v}(\varphi; \tau)}{\hat{u}_0(\varphi)}\right), \tag{23}$$

where $\sigma = c\tau/h \ll 1$. The numerical dispersion error may be identified based on deviations of the real part of Φ from the ideal φ behavior, and the diffusion error based on the imaginary part of Φ. Despite limitations discussed in the original reference, EDR is capable of accounting to leading order for nonlinear effects contained in shock–capturing schemes. The results of the EDR analysis are reported in Fig. 5, where we also show for reference the distributions of the spectral transfer function for the underlying CU7 and WENO7 schemes. As expected, the EDR obtained with Harten's, Jameson's, Visbal's and Multi-resolution sensors are very similar to the baseline WENO7 scheme, which confirms that those sensors cannot effectively distinguish shocks from sinusoidal waves. Hill's and Movahed's sensors perform slightly better, as they yield less numerical dispersion and diffusion compared with WENO7 for well-resolved waves. Perhaps unexpectedly based on what previously observed, Ren's sensor yields substantial advantage here, as its EDR is not far from the baseline CU7 scheme. This is probably due to the fact (see Fig. 2) that this sensor only applies WENO to a limited number of points in a wave. The new sensor shows the best results in terms of EDR, and deviations from the CU7 quasi-spectral behavior only show up when fewer than three points-per-wavelength are used, thus further supporting high quality and resolution ability. Minor differences are found when different of the parameter exponent θ are used.

As a second step, the performance of compact/WENO algorithms with various shock sensors is analyzed by carrying out numerical simulations of the test case originally considered by Casper and Carpenter [38], consisting in the linear propagation of sinusoidal disturbances in a medium with abrupt change of the speed of sound, and modeling the refraction of sound waves at the interface of solid media with different acoustic impedances. We have considered a grid spacing such that the reduced wavenumber past the shock is $\varphi_2 = w_2 h \approx 1.12$, corresponding to about **6** points per wavelength. This particular value was selected for being the highest wavenumber for which local application of WENO7 at the shock does not yield large numerical damping [39]. Fig. 6 shows numerical solutions obtained with various shock sensors, again assuming the threshold values listed in Table 3. As appropriate, all the sensors activate the WENO flux around the shock location, thus enabling shock capturing. However, Harten's, Jameson's and Visbal's sensors also mark points away from the shock as troubled zones. As a result, the numerical solution exhibits severe damping of waves past the shock. Ren's sensor also has sub-optimal performance, interpreting poorly resolved waves as discontinuities. All other sensors are found to perform correctly, the WENO switch being only turned on around the shock position, although it should be noted that Multi-resolution, Hill's and Movahed's sensors are probably marking as troublesome a bit too wide a region. Our novel shock sensor precisely marks the shock position, and WENO is activated at barely four points, thus minimizing numerical dissipation and dispersion in smooth regions. Careful inspection of the numerical results does in fact show slight phase lead for waves

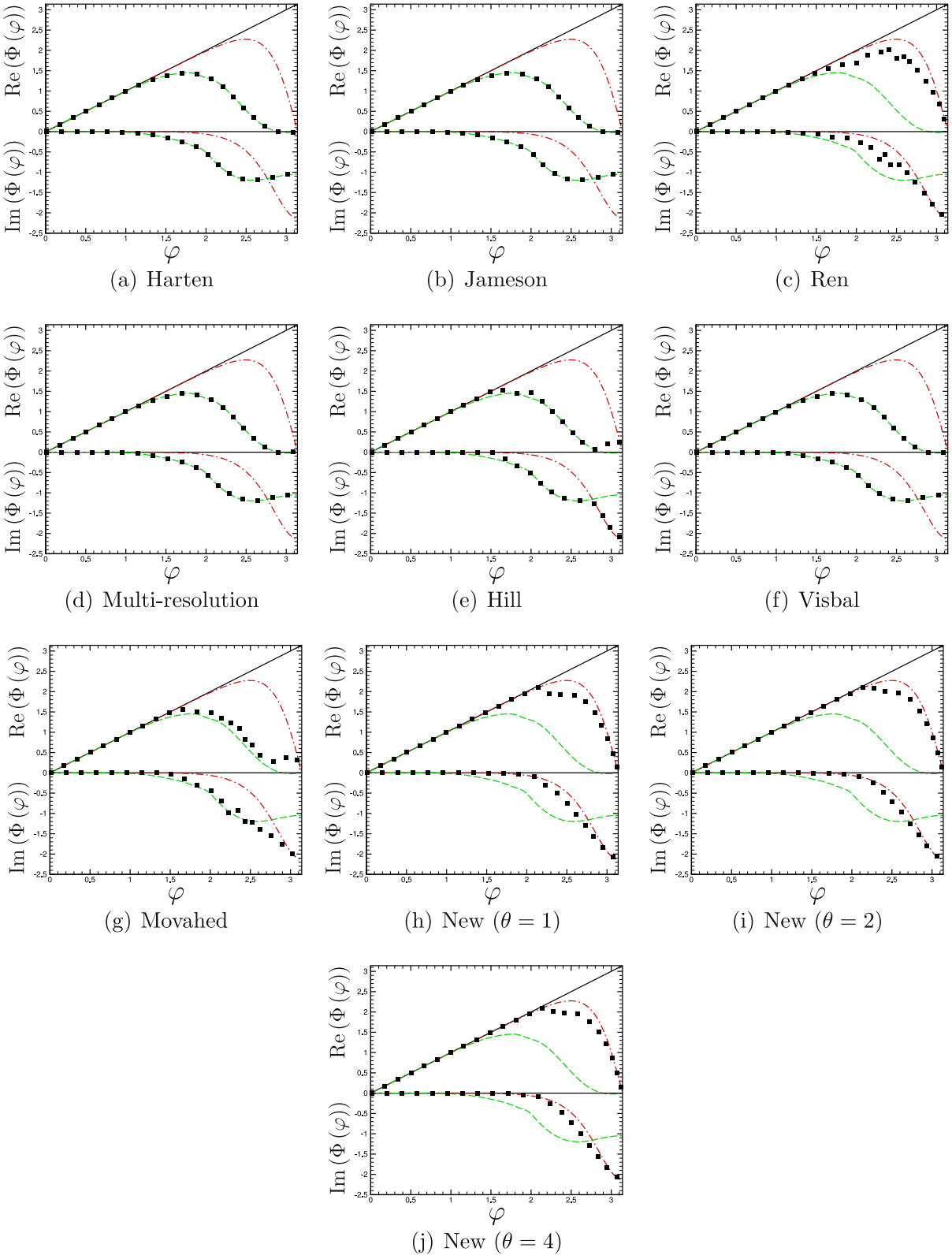


Fig. 5. Empirical dispersion relation (EDR) of hybrid CU7/WENO7 scheme with different shock sensors. The real and the imaginary parts of the spectral transfer function are shown in the top and bottom parts of each panel, respectively. The reference EDR for the CU7 (Dash-dotted lines) and WENO7 (dashed lines) schemes are also shown, as well as the ideal spectral behavior $\Phi = \varphi$ (solid lines).

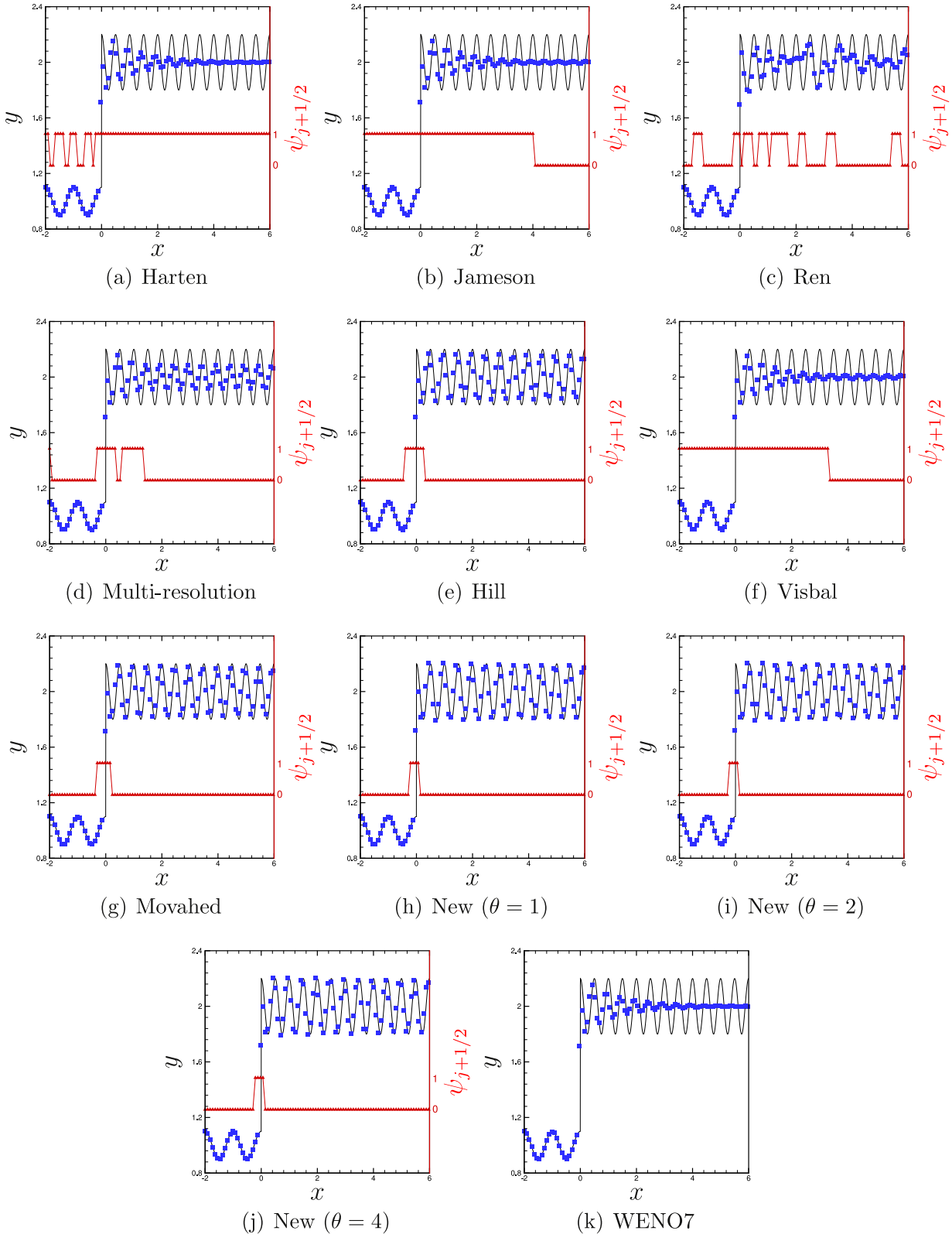


Fig. 6. Numerical results of sound wave refraction test case with different shock sensors, using the threshold values given in Table 3. Ninety grid points are used, corresponding to about six points-per-wavelength downstream of the shock. The numerical solution is shown with square symbols, and compared with the exact solution (solid lines). The triangles indicate local values of the shock switch $\psi_{j+1/2}$.

past the shock as compared to the exact solution, which is a consequence of local application of WENO7 at the shock [39].

5. Application to Euler equations

We now discuss application of the shock sensors previously introduced to two test cases involving the Euler equations, which were extensively studied in the literature. In order to adapt the compact/WENO algorithm previously described in the scalar case, upwinding is applied separately to each characteristic field, upon Lax-Friedrichs flux splitting [31]. With regard to the shock sensor, we only use information from the density field, which allows substantial computational savings [2]. The same threshold values ϵ_{SC} given in Table 3 used for the scalar analysis, are here retained. Calculations have been carried out at constant CFL number, $\sigma = 0.5$.

5.1. One-dimensional shock/entropy wave interaction

The Shu-Osher test case [40] is widely used in the literature as a benchmark to test the capability of numerical methods to resolve small-scale features while preserving the necessary robustness to capture shock waves with moderate-to-strong intensity. The one-dimensional Euler equations are solved with the initial conditions

$$(\rho, u, p) = \begin{cases} (3.85714, \quad 2.629369, \quad 10.33333), & x < -4 \\ (1 + 0.2\sin(5x), \quad 0, \quad 1), & x \geq -4 \end{cases}, \tag{24}$$

to reproduce the unsteady interaction of a $Ma = 3$ shock wave with a sine entropy wave carrying density fluctuations of amplitude $\Delta\rho = 0.2$. The numerical solution is computed in the space interval $x \in [-5, 5]$, and integrated in time up to $t = 1.8$. Zero time variation of the conservative variables is specified at the left and right boundaries, which is appropriate for the time interval under interest. This test case is quite challenging because the solution includes smooth regions with a rich structure consisting of receding acoustic waves (which steepen and give rise to shocklets) as well as entropy waves propagating downstream of the shock. The density fields computed with 121 and 161 grid points are shown in Figs. 7 and 8, respectively. These corresponds to really marginal resolution cases, for which the entropy wave downstream of the shock is resolved with as few as three and five grid points, respectively. Notably, this resolution is much less than commonly shown in the literature (typically, 201 grid points are used), and classical high-order schemes such as WENO7 fail dramatically (see panels (j) of the figures). At this marginal resolution, most sensors mark most grid points as critical, also including zones with propagating waves, and as a result the density distributions are not very different than pure WENO7. Substantial improvement is obtained with Ren's sensor, Hill's and Movahed's sensors to even greater extent. The latter two sensors in fact correctly do not identify entropy waves as shocks, but tend to be over-sensitive to steepening acoustic waves. In this respect, the new sensor tends to yield better results as only fully steepened fronts are perceived as shock. Close-up view of the density right past the shock provided in the figure insets further supports greater resolving capability of the new sensor over other high-resolution sensors, in that the amplitude of density oscillations is correctly preserved. This may be explained by noting that, while no point is handled in the WENO mode with any of the high-resolution sensors at the final time, WENO7 is occasionally activated at certain time instants when Hill's or Movahed's sensors are used, which explains the slightly worse performance as far as wave propagation is concerned. As with the basic scalar tests, no significant effect results from change of the θ parameter in the new sensor. Most important, the resolution improvement offered by the new sensor is more significant on the coarser mesh.

In order to illustrate that the analysis of shock sensors carried out so far is of sufficiently general fetch, we have repeated the analysis of the Shu-Osher test case by replacing WENO7 with a targeted ENO scheme [TENO, [29,41–43]], as a shock-capturing bulding block, whereas the CU7 is retained as baseline linear scheme. In particular, a seventh-order version of TENO (referred to as TENO7) is considered here, whereby the original WENO weights are replaced as suggested by Borges et al. [26]. The results obtained with various sensors are shown in Fig. 9, along with the shock-capturing thresholds which have been used. The figure supports generally higher resolution properties of TENO7 as compared to classical WENO7 (compare panels (k) of Figs. 8 and 9). The figure also suggests slightly better performance in the case hybrid computations, also on account of larger allowed shock-capturing thresholds. Superiority of the new class of sensors herein developed in also confirmed when TENO is used instead of WENO.

5.2. Two-dimensional Riemann problem

In this two-dimensional example, corresponding to Test Case 3 of Lax and Liu [44], a Riemann problem is considered in a unit square with the following initial conditions

$$(\rho, u, v, p) = \begin{cases} (1.5, \quad 0, \quad 0, \quad 1.5) & \text{if } 0.8 \leq x \leq 1, \ 0.8 \leq y \leq 1, \\ (0.5323, \quad 1.206, \quad 0, \quad 0.3) & \text{if } 0 \leq x \leq 0.8, \ 0.8 \leq y \leq 1, \\ (0.138, \quad 1.206, \quad 1.206, \quad 0.029) & \text{if } 0 \leq x \leq 0.8, \ 0 \leq y \leq 0.8, \\ (0.5323, \quad 0, \quad 1.206, \quad 0.3) & \text{if } 0.8 \leq x \leq 1, \ 0 \leq y \leq 0.8, \end{cases} \tag{25}$$

yielding a system of four shocks interacting with each other, and generating a complex pattern of regular and Mach reflections with associated slip lines originating at the triple points. Interaction of the two stronger slip lines gives rise to a mushroom-shaped jet head pointing toward the bottom-left corner. The quality of a numerical scheme mainly resides in the capability to capture more and more eddies resulting from Kelvin-Helmholtz instability of the slip lines, and their complex interaction in the head of the jet. The test is made even more challenging for hybrid algorithms by the presence of a Mach stem embedded in the jet head associated with pressure recovery from supersonic state. The computed density fields are presented in the left columns of Fig. 10 on a 300^2 mesh, whereas the right columns show zones where WENO is activated, as controlled by the sensor threshold values given in Table 3. The solutions are computed up to $t = 0.8$. As in the one-dimensional test, the shock sensors are evaluated on the basis of the density field. At this mesh resolution, the basic WENO7 scheme (panel (j)) captures the main flow features, including shocks, slip lines, and jet without any hint of Kelvin-Helmholtz roll-up, here inhibited by excessive numerical diffusion. Discrete vortices are found to form in the jet wake when classical sensors are used (panels (a), (b), (e)). Finer details are resolved when high-resolution sensors are used, and earlier formation of discrete vortices along the slip lines is observed. Additional details in the starting vortex are also observed when the new sensor is used, especially with $\theta = 1$.

Slight asymmetry of the jet structure should be noticed in some numerical simulations carried out with WENO-based sensors. This is caused by nearly random switch between WENO and compact discretization which may even be triggered by inevitable round-off error, and which here is not inhibited by physical viscosity and numerical diffusion. Further, as may be seen in the right panels of the figure, some zones which are expected to be 'smooth' in fact contain scattered points which are identified as 'critical', which is inevitable especially as the simulation is inviscid. With the exception of multi-resolution sensor, the new sensor is least affected by this phenomenon than other sensors. The presence of scattered

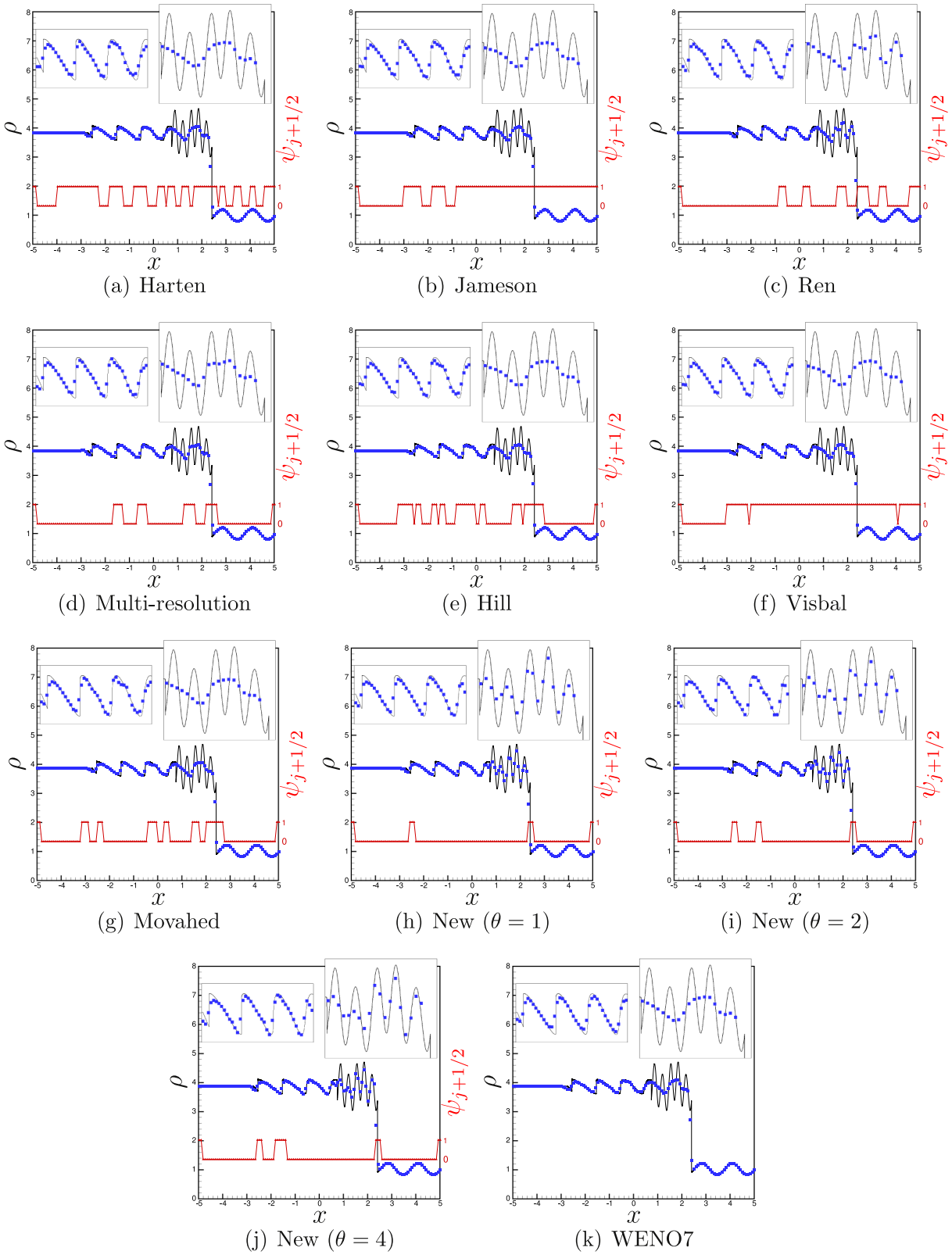


Fig. 7. Shu-Osher test case on a 120 cells mesh: distribution of density (square symbols) and shock switch (triangles). A reference WENO7 solution obtained with 40,000 cells is shock with a thick solid line.

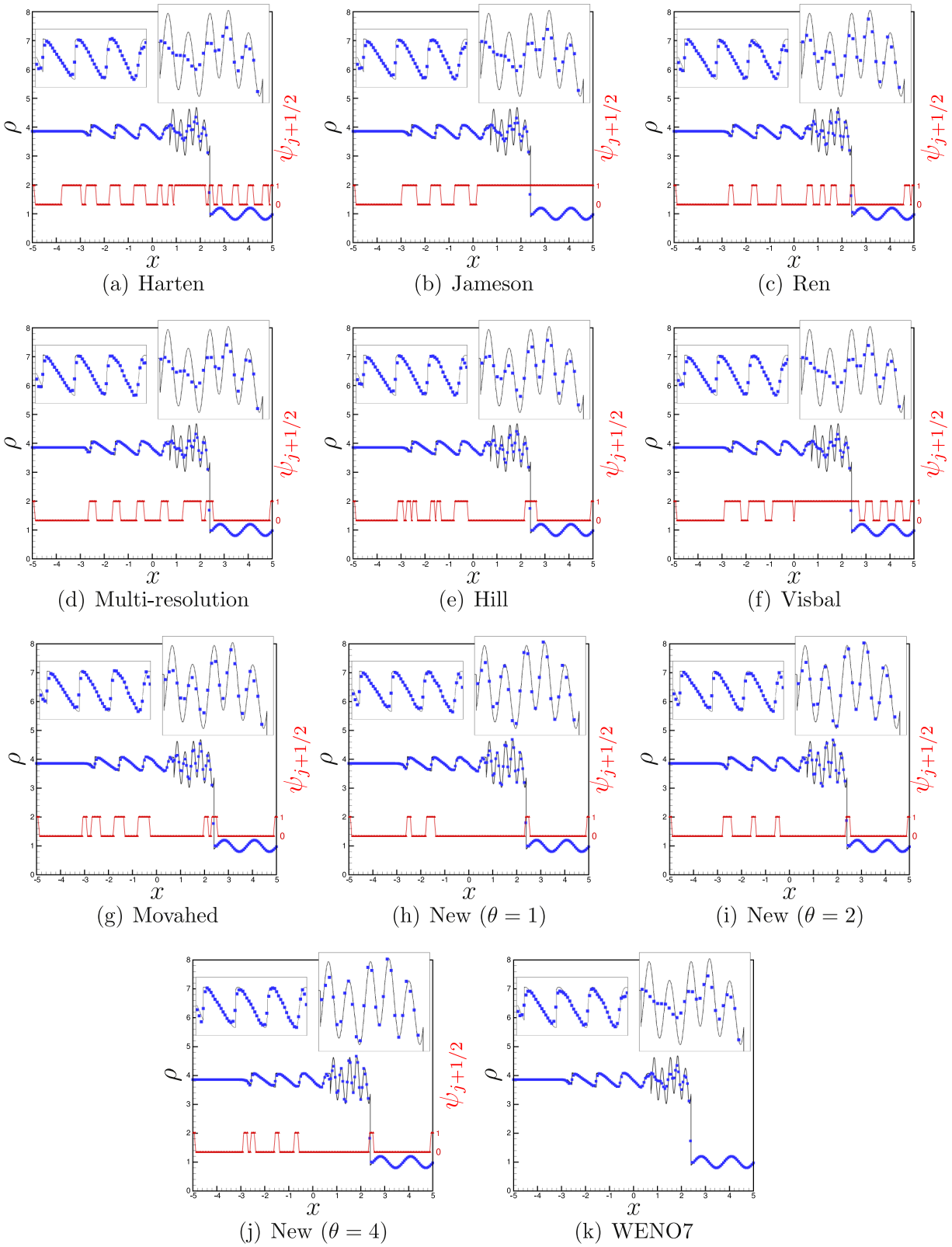
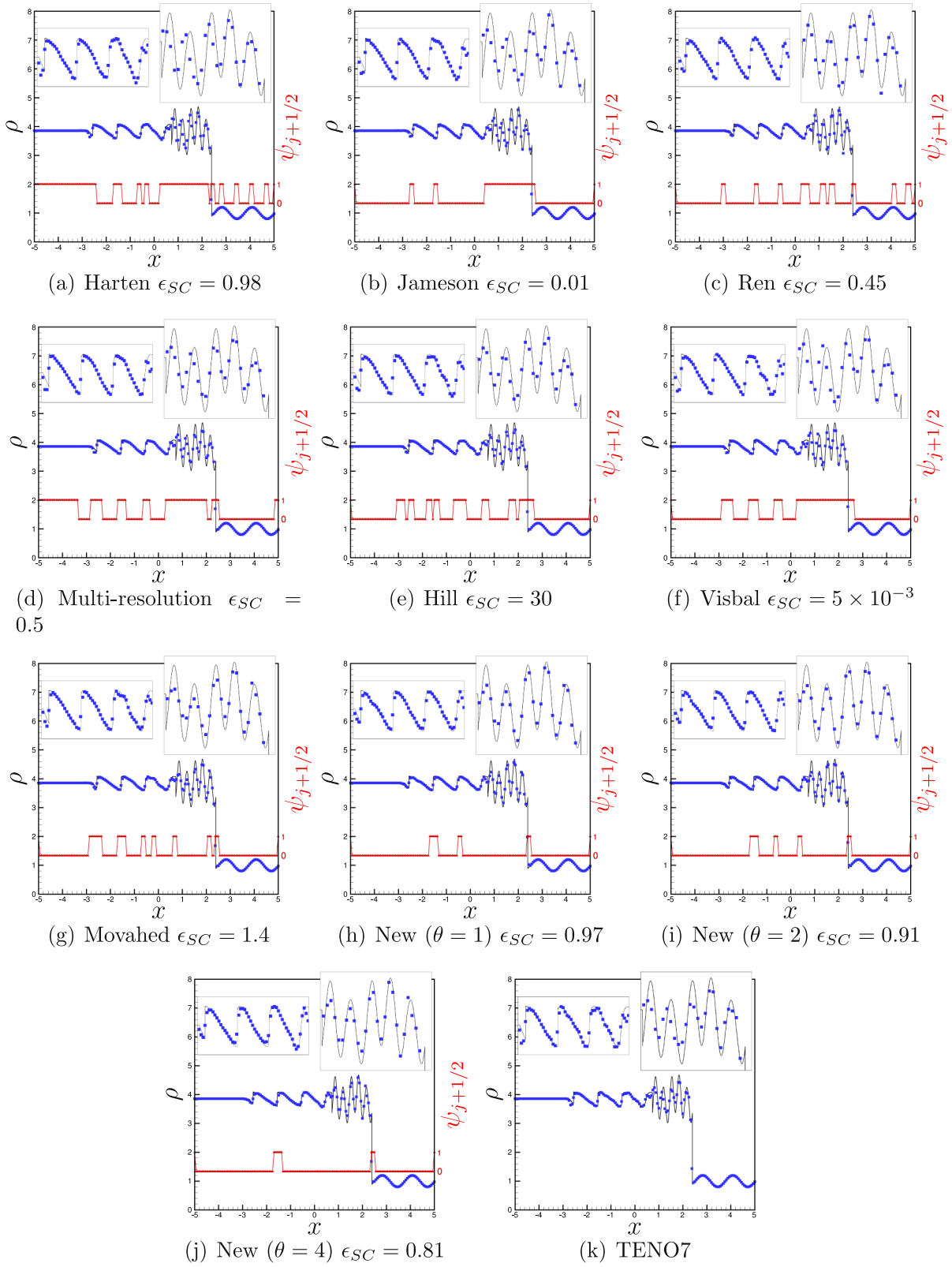


Fig. 8. Shu-Osher test case on a 160 cells mesh: distribution of density (square symbols) and shock switch (triangles). A reference WENO7 solution obtained with 40,000 cells is shock with a thick solid line.



(a) Harten $\epsilon_{SC} = 0.98$ (b) Jameson $\epsilon_{SC} = 0.01$ (c) Ren $\epsilon_{SC} = 0.45$

(d) Multi-resolution $\epsilon_{SC} = 0.5$ (e) Hill $\epsilon_{SC} = 30$ (f) Visbal $\epsilon_{SC} = 5 \times 10^{-3}$

(g) Movahed $\epsilon_{SC} = 1.4$ (h) New $(\theta = 1)$ $\epsilon_{SC} = 0.97$ (i) New $(\theta = 2)$ $\epsilon_{SC} = 0.91$

(j) New $(\theta = 4)$ $\epsilon_{SC} = 0.81$ (k) TENO7

Fig. 9. Shu-Osher test case on a 160 cells mesh using hybrid CU7/TENO7 scheme: distribution of density (square symbols) and shock switch (triangles). A reference WENO7 solution obtained with 40,000 cells is shock with a thick solid line.

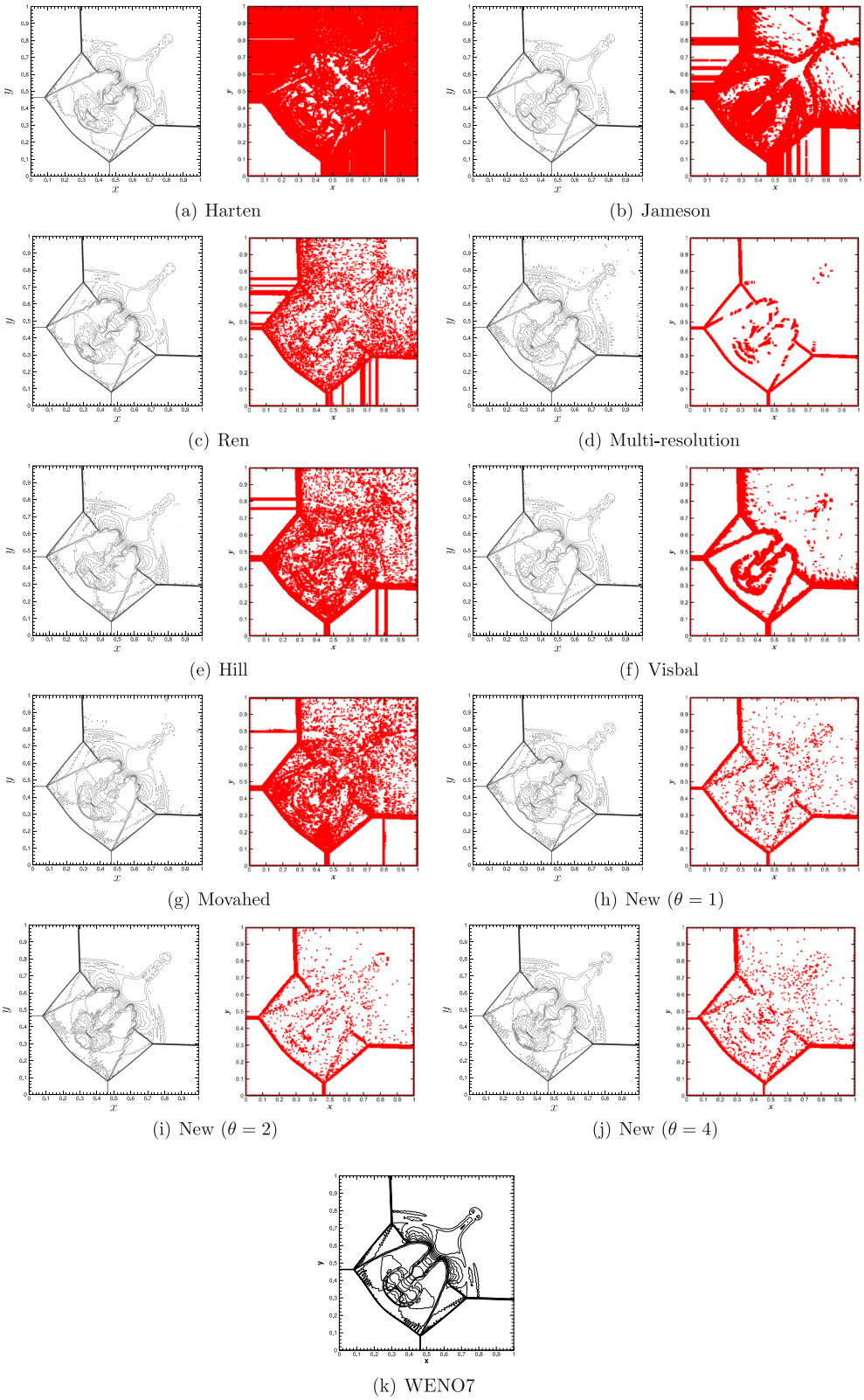


Fig. 10. Numerical simulation of Test Case 3 of Lax and Liu [44] with various shock sensors. The unit square is covered with a 300^2 cells mesh, and the solution is shown at $t = 0.8$. Left panels show density contours (fifteen levels, from 0.3 to 1.6), and right panels highlight zones where shock-capturing is activated.

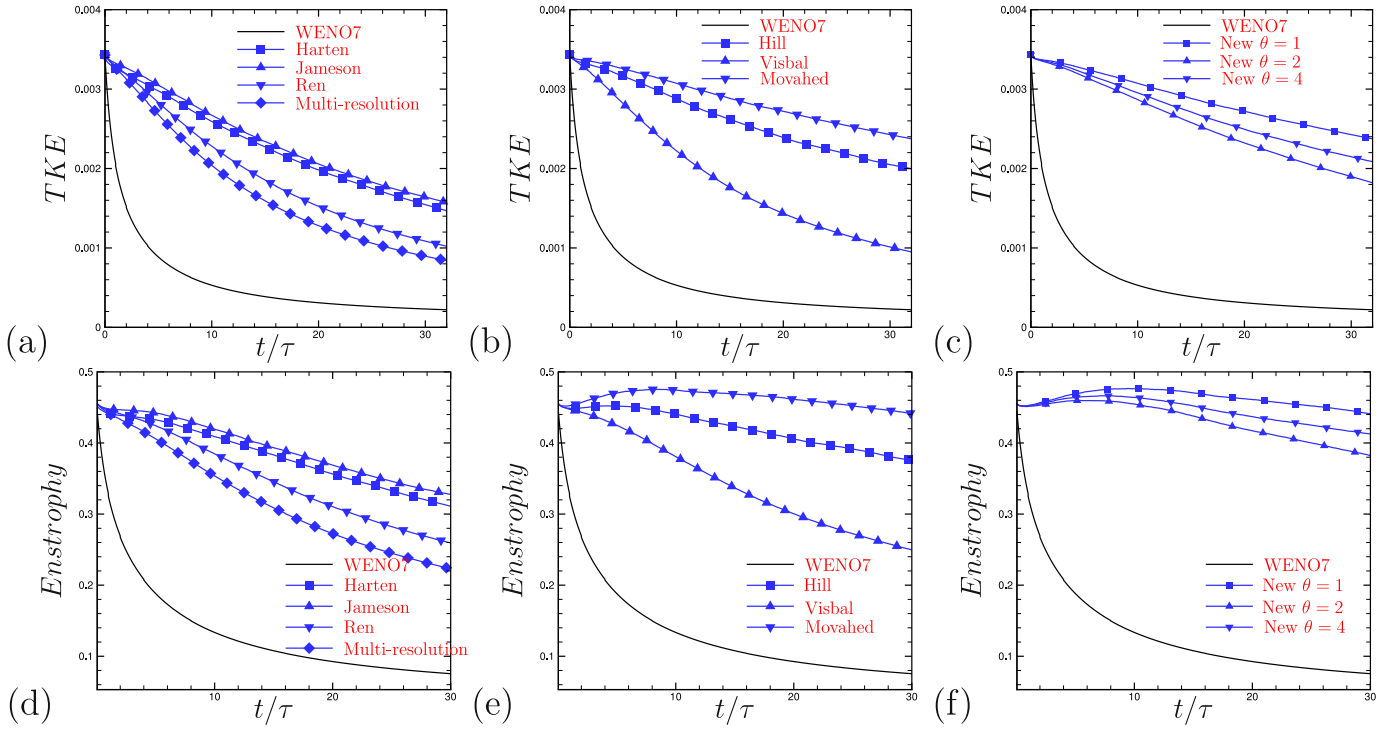


Fig. 11. Time evolution of kinetic energy (a–c) and enstrophy (d–f) for isotropic turbulence at zero viscosity ($M_{t0} = 0.07$), for classical and multi-resolution sensors (a,d), WENO-based sensors (b,e), and new sensors (c,f). τ is the eddy turnover time.

zones to be handled in the WENO mode has implications of the overall computational efficiency of the algorithm, as it does not allow effective use of the CPU vector hardware. Nevertheless, hybrid schemes are found to work quite well in practice, and they have for instance been used for large-scale simulations of shock/turbulent boundary layer interactions [45,46].

5.3. Euler turbulence

To test the performance of hybrid schemes with different shock sensors in more realistic cases of flow turbulence we have carried out a series of numerical simulations of (unforced) isotropic compressible turbulence in a $(2\pi)^3$ periodic box at zero molecular viscosity, as proposed by Honein and Moin [47]. The calculations are initialized with a field of synthetic isotropic solenoidal turbulence with assigned wavenumber spectrum

$$E(k) = A\,(k/k_0)^4\,\exp\left[-2(k/k_0)^2\right], \tag{26}$$

peaking at $k_0 = 6$. The constant A is adjusted in such a way that the initial turbulent Mach number $M_t = u_{rms}/\bar{c}$ (where the prime indicates the r.m.s. value, and $\bar{c}$ is the mean speed of sound) is set to a desired value. Fluctuations of temperature and density are initially set to zero. Under these weakly compressible conditions, energy should be nearly conserved, and an equipartition k^2 energy spectrum should develops [48], with associated growth of the enstrophy. Numerical instability is then expected in the absence of physical viscosity, which is prevented by numerical diffusion. Results of numerical simulations are shown in Fig. 11 for initial turbulent Mach number $M_{t0} = 0.07$, where kinetic energy and enstrophy evolution are reported. As expected, built-in numerical diffusion causes decrease of kinetic energy in time, however at much different rate across schemes. In fact, all hybrid schemes yield much milder decrease than classical WENO7. As also noted in the previous analysis, WENO-based sensors are found to behave better than classical ones, and the results obtained with Movahed's and the new sensor (with $\theta = 1$) are quite similar. Indeed, recalling the analysis reported in Fig. 4, the two sensors are expected to perform similarly in the limit of poorly resolved waves. Similar conclusions may be made with respect to the enstrophy, which only exhibits some slight growth is Movahed's and the new sensor are used, whereas the formation of small scales with all other sensors is prevented by numerical diffusion.

6. Conclusion

We have presented a systematic framework for evaluating the performance of shock sensors in wavenumber space applied to hybrid compact/WENO algorithms. In this class of schemes shock-capturing WENO reconstruction is used to locally replace a baseline compact upwind reconstruction made to maximize the resolving efficiency in smooth parts of the flow, the switch being controlled by the local value of the sensor function. Several shock sensors have been considered, including classical ones, based on low-order derivative of the numerical solution, as well as modern ones, generally based on WENO smoothness indicators or weights. A critical issue in practical use of shock sensors is the identification of a suitable threshold for switch activation. This is a non-trivial matter of compromise between the competing needs to maintain the shock-capturing capability (which suggests low values), and to retain good resolution of waves (which prompts larger values). In this paper we have followed the pragmatic approach of preliminarily evaluating the largest possible threshold value for each sensor compatible with algorithmic robustness. Although different sensor thresholds should be used in general for different combinations of baseline schemes, the overall sensor evaluation procedure would not change. Of course, selecting lower thresholds is possible, but that would spoil the overall scheme performance due to excessive numerical diffusion. Static and dynamic tests of compact/WENO schemes equipped with various sensors have been carried out to establish the respective resolving efficiency, namely the range of

Table 5
Estimated CPU time (normalized with respect to Harten's sensor) for compact/WENO schemes with different shock sensors.

Sensor	Harten	Jameson	Ren	Multi-resolution	Hill	Visbal	Movahed	New
CPU	1	0.941	0.947	1.301	1.413	1.341	1.348	1.344

wavenumbers for which no grid point is marked as critical, hence the underlying compact scheme doesn't suffer from contamination from WENO larger numerical diffusion. The resolving efficiency of classical sensors is found to be very small, whereas WENO-based sensors are found to be capable of resolving waves with barely up to four points-per-wavelength. A new sensor is designed and tested, based of the norm of the normalized deviation of WENO weights from their optimal value, which is capable of discerning waves with as few three points-per-wavelength. As a result, the spectral transfer function of the resulting hybrid compact/WENO scheme is significantly improved over other sensors, and it is found to yield reduced numerical diffusion in wave propagation problems past shocks, also including the case of the Euler equations. Favourable performance of WENO-based sensors, and especially of the new sensor, is confirmed in multi-dimensional Euler calculations including a mixture of shocks and smooth features. An overarching issue in practical use of high-resolution compact-WENO schemes is their computational cost. This is especially true for WENO-based sensors, which are found to have excellent resolution in wavenumber space, but which require evaluation of the WENO weight everywhere, and not only at points where WENO is actually applied. Some mitigation of the necessary computational cost is provided by use of WENO weights based on the density field only, rather than on each primitive variable or characteristic field. For reference purposes, in Table 5 we report a list of the typical CPU time needed to carry out the two-dimensional Euler computations. As one can see, the typical total overhead for implementation of WENO-based sensors is about 30–40% with respect to classical sensors, a price which is typically well paid off by the large improvement in the resolving power of the resulting scheme. We emphasize that although the analysis of shock sensors is here mainly carried out for a particular combination of baseline schemes, namely *CU*7 and *WENO*7, and for shock sensors based on the latter, the approach has general validity, hence it can in principle be applied to any kind of sensor and hybrid scheme [49,50]. For instance, modern shock sensors based on multi-resolution wavelet analysis and TENO shock-capturing building blocks have also been tested. However, our intent here is not to endorse any specific hybrid scheme or another, but rather to build a comprehensive framework which enables cross-comparison of different options. As a final remark, we should point out that, especially in 2D test case previously presented, some weak discontinuities are not detected to be discontinuous. This of course improves the resolution, at the expense of making the solution more oscillatory. This raises the fundamental question of when a flow region should be actually tagged as a 'discontinuity'. Here, we have made the pragmatic choice that a flow zone should not be treated as a discontinuity unless it yields numerical divergence. This is of course far from being satisfactory from a theoretical standpoint, and additional research is probably appropriate to tackle this long-standing fundamental issue. Equally important is whether the results herein reported directly carry on to realistic flows. Numerical simulations of isotropic compressible turbulence at zero viscosity generally supports inferences made on simple 1D and 2D test cases. The eventual goal of the on-going research is the high-fidelity simulation of compressible turbulence with shocks, either explicit LES or DNS. In those cases viscous terms and/or subgrid scale models should also be used, whose magnitude is problem and Reynolds number dependent, and whose discretization may also affect the behavior of flow solvers. These important issues are left for future studies.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organi-zations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

CRediT authorship contribution statement

Guo-Yan Zhao: Data curation, Formal analysis, Investigation, Writing - original draft. **Ming-Bo Sun:** Funding acquisition, Supervision. **Sergio Pirozzoli:** Software, Validation, Writing - review & editing.

Acknowledgments

This work is supported by National Natural Science Foundation of China under Grants nos. 11522222 and 11472305, Innovative Sustentation Fund for Excellent Ph.D. Students in Hunan province (No. CX2016B003), and Chinese Scholarship Council (CSC) (No. 201703170277) during a visit of Guo-Yan Zhao in Rome.

References

[1] Adams NA, Shariff K. A high-resolution hybrid compact-ENO scheme for shock--turbulence interaction problems. J Comput Phys 1996;127(1):27–51.
[2] Pirozzoli S. Conservative hybrid compact-WENO schemes for shock-turbulence interaction. J Comput Phys 2002;178(1):81–117.
[3] Harten A. The artificial compression method for computation of shocks and contact discontinuities. III. Self-adjusting hybrid schemes. Math Comput 1978;32(142):363–89.
[4] Jameson A, Schmidt W, Turkel E. Numerical solution of the Euler equations by finite volume methods using Runge–Kutta time stepping schemes. In: 14th fluid and plasma dynamics conference; 1981. p. 1259.
[5] Ren Y-X, Zhang H, et al. A characteristic-wise hybrid compact-WENO scheme for solving hyperbolic conservation laws. J Comput Phys 2003;192(2):365–86.
[6] Bogey C, De Cacqueray N, Bailly C. A shock-capturing methodology based on adaptative spatial filtering for high-order non-linear computations. J Comput Phys 2009;228(5):1447–65.
[7] Oliveira M, Lu P, Liu X, Liu C. A new shock/discontinuity detector. Inter J Comput Math 2010;87(13):3063–78.
[8] Sjögreen B, Yee HC. Multiresolution wavelet based adaptive numerical dissipation control for shock-turbulence computations. Res Inst for Adv Comput Sci 2001;1:1–43.
[9] Sjögreen B, Yee HC. Multiresolution wavelet based adaptive numerical dissipation control for high order methods. J Sci Comput 2004;20(2):211–55.
[10] Yee HC, Sjögreen B. Efficient low dissipative high order schemes for multiscale MHD flows, II: minimization of b numerical error. J Sci Comput 2006;29(1):115–64.
[11] Yee HC, Sjögreen B. Development of low dissipative high order filter schemes for multiscale Navier–Stokes/MHD systems. J Comput Phys 2007;225(1):910–34.
[12] Yee HC, Sjögreen B. Adaptive filtering and limiting in compact high order methods for multiscale gas dynamics and MHD systems. Comput Fluids 2008;37(5):593–619.
[13] Kotov DV, Yee HC, Wray AA, Sjögreen B, Kritsuk AG. Numerical dissipation control in high order shock-capturing schemes for LES of low speed flows. J Comput Phys 2016;307:189–202.
[14] Kotov DV, Yee HC, Wray AA, Hadjadj A, Sjögreen B. High order numerical methods for the dynamic SGS model of turbulent flows with shocks. Commun Comput Phys 2016;19(2):273–300.
[15] Hill DJ, Pullin DI. Hybrid tuned center-difference-WENO method for large eddy simulations in the presence of strong shocks. J Comput Phys 2004;194(2):435–50.

[16] Visbal M, Gaitonde D. Shock capturing using compact-differencing-based methods. In: 43rd AIAA aerospace science meeting and exhibit; 2005. p. 1265.
[17] Movahed P, Johnsen E. A solution-adaptive method for efficient compressible multifluid simulations, with application to the Richtmyer–Meshkov instability. J Comput Phys 2013;239:166–86.
[18] Fu H, Wang Z, Yan Y, Liu C. Modified weighted compact scheme with global weights for shock capturing. Comput Fluids 2014;96:165–76.
[19] Peng J, Shen Y. A novel weighting switch function for uniformly high-order hybrid shock-capturing schemes. Int J Numer Methods Fluids 2017;83(9):681–703.
[20] Ducros F, Ferrand V, Nicoud F, Weber C, Darracq D, Gacherieu C, Poinsot T. Large-eddy simulation of the shock/turbulence interaction. J Comput Phys 1999;152(2):517–49.
[21] Larsson J, Lele SK, Moin P. Effect of numerical dissipation on the predicted spectra for compressible turbulence; 2007 Annu Res Briefs:47–57.
[22] White J, Baurle R, Fisher T, Quinlan J, Black W. Low-dissipation advection schemes designed for large eddy simulations of hypersonic propulsion systems. In: 48th AIAA/ASME/SAE/ASEE joint propulsion conference & exhibit; 2012. p. 4263.
[23] Lo S-C, Blaisdell GA, Lyrintzis AS. High-order shock capturing schemes for turbulence calculations. Int J Numer Methods Fluids 2010;62(5):473–98.
[24] Fang J, Li Z, Lu L. An optimized low-dissipation monotonicity-preserving scheme for numerical simulations of high-speed turbulent flows. J Sci Comput 2013;56(1):67–95.
[25] Henrick AK, Aslam TD, Powers JM. Mapped weighted essentially non-oscillatory schemes: achieving optimal order near critical points. J Comput Phys 2005;207:542–67.
[26] Borges R, Carmona M, Costa B, Don WS. An improved weighted essentially non-oscillatory scheme for hyperbolic conservation laws. J Comput Phys 2008;227:3191–211.
[27] Hu XY, Wang Q, Adams NA. An adaptive central-upwind weighted essentially non-oscillatory scheme. J Comput Phys 2010;229(23):8952–65.
[28] Fan P. High order weighted essentially nonoscillatory WENO-η schemes for hyperbolic conservation laws. J Comput Phys 2014;269:355–85.
[29] Fu L, Hu XY, Adams NA. A family of high-order targeted ENO schemes for compressible-fluid simulations. J Comput Phys 2016;305:333–59.
[30] Zhang H, Zhang F, Xu C. Towards optimal high-order compact schemes for simulating compressible flows. Appl Math Comput 2019;355:221–37.
[31] Jiang G-S, Shu C-W. Efficient implementation of weighted ENO schemes. J Comput Phys 1996;126(1):202–28.
[32] Lele SK. Compact finite difference schemes with spectral-like resolution. J Comput Phys 1992;103(1):16–42.
[33] Larsson J, Gustafsson B. Stability criteria for hybrid difference methods. J Comput Phys 2008;227(5):2886–98.
[34] Balsara DS, Shu C-W. Monotonicity preserving weighted essentially non-oscillatory schemes with increasingly high order of accuracy. J Comput Phys 2000;160(2):405–52.
[35] Mallat S, Zhong S. Characterization of signals from multiscale edges. IEEE Trans Pattern Anal Mach Intell 1992;14(7):710–32.
[36] Harten A. Multiresolution algorithms for the numerical solution of hyperbolic conservation laws. Commun Pur Appl Math 1995;48(12):1305–42.
[37] Pirozzoli S. On the spectral properties of shock-capturing schemes. J Comput Phys 2006;219(2):489–97.
[38] Casper J, Carpenter MH. Computational considerations for the simulation of shock-induced sound. J Sci Comput 1998;19(3):813–28.
[39] Zhao G-Y, Sun M-B, Memmolo A, Pirozzoli S. A general framework for the evaluation of shock-capturing schemes. J Comput Phys 2019;376:924–36.
[40] Shu C-W, Osher S. Efficient implementation of essentially non-oscillatory shock-capturing schemes, II. In: Upwind and high-resolution schemes. Springer; 1989. p. 328–74.
[41] Fu L, Hu XY, Adams NA. Targeted ENO schemes with tailored resolution property for hyperbolic conservation laws. J Comput Phys 2017;349:97–121.
[42] Fu L, Hu XY, Adams NA. Improved five-and six-point targeted essentially nonoscillatory schemes with adaptive dissipation. AIAA J 2018;57(3):1143–58.
[43] Fu L, Tang Q. High-order low-dissipation targeted ENO schemes for ideal magnetohydrodynamics. J Sci Comput 2019;80(1):692–716.
[44] Lax PD, Liu X-D. Solution of two-dimensional Riemann problems of gas dynamics by positive schemes. J Sci Comput 1998;19(2):319–40.
[45] Pirozzoli S. Generalized conservative approximations of split convective derivative operators. J Comput Phys 2010;229(19):7180–90.
[46] Pirozzoli S. Numerical methods for high-speed flows. Ann Rev Fluid Mech 2011;43:163–94.
[47] Honein AE, Moin P. Higher entropy conservation and numerical stability of compressible turbulence simulations. J Comput Phys 2004;201(2):531–45.
[48] Kraichnan RH. On the statistical mechanics of an adiabatically compressible fluid. J Acoust Soc Am 1955;27(3):438–41.
[49] Zhao G-Y, Sun M-B, Xie S-B, Wang H-B. Numerical dissipation control in an adaptive WCNS with a new smoothness indicator. Appl Math Comput 2018;330:239–53.
[50] Zhao G-Y, Sun M-B, Mei Y, Li L, Wang H-B, Li G-X, Liu Y, Sun Y-C, Liang C-H. An efficient adaptive central-upwind WENO-CU6 numerical scheme with a new sensor. J Sci Comput 2019;81(2):649–70.